Introduction to Credit Risk and Applications in Python

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Lecture Outline

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Introduction

The Illusion of Stability

- Most of the time, portfolios appear stable.
- Observed default rates are low.
- Losses cluster around small expected values.

Why Study Credit Risk?

If Defaults Are Rare, Where Does Systemic Risk Come From?

- **Dependence —** Defaults do not occur independently.
- **Factor structure —** Common economic drivers create clustering.
- **Model assumptions —** Distributional choices shape outcomes.
- **Tail amplification —** Small structural effects become large in extremes.

Dependence and Tail Amplification: A Simulation

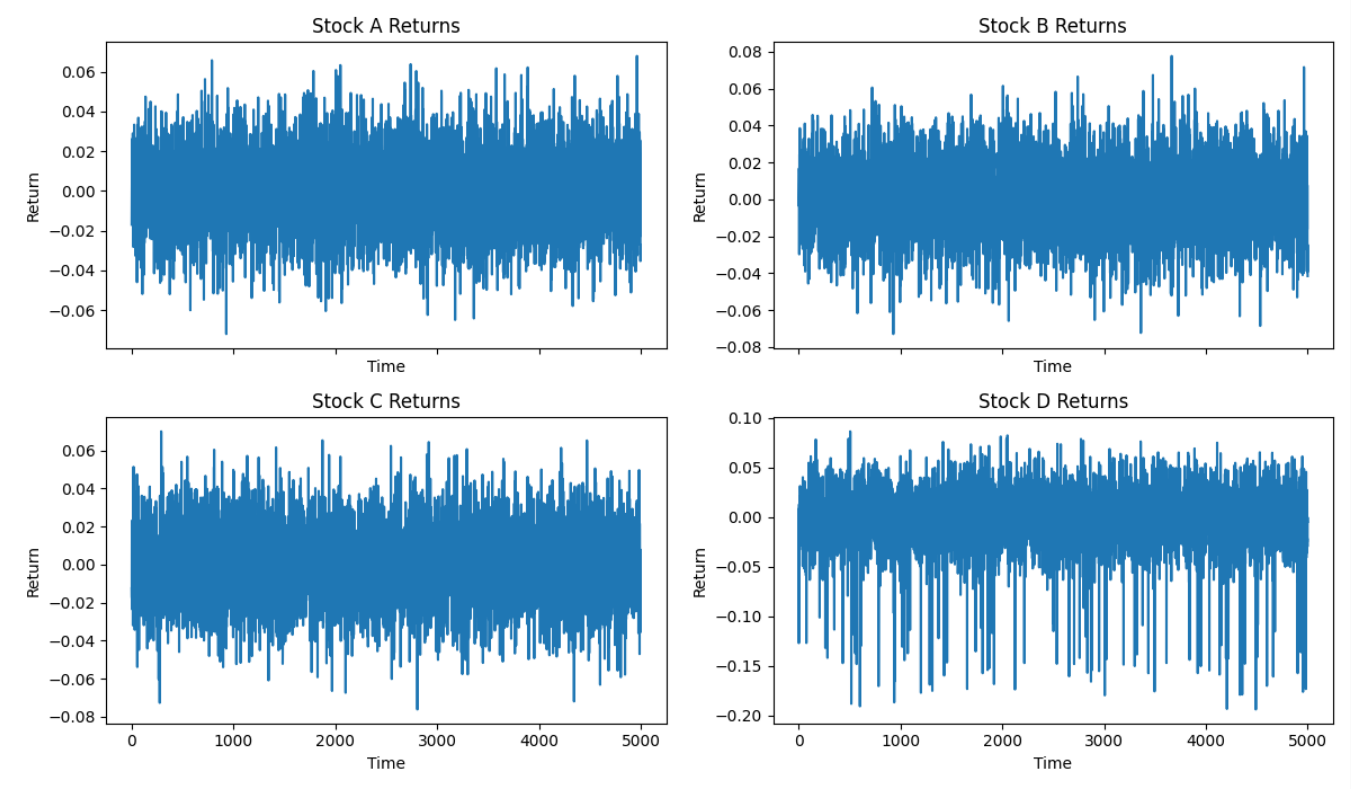


Figure 1: Simulated Multivariate Returns with Correlation and Tail Shock

Uncomfortable Facts

- **Fact 1 — Credit risk is usually invisible in calm periods.**
- **Fact 2 — Correlation drives extreme portfolio losses.**
- **Fact 3 — Model assumptions determine measured capital.**

From Stability to Systemic Risk

- Risk is not about average losses.
- Risk is about rare joint events.
- Banks do not fail because a single borrower defaults.
- Banks fail when many exposures move together.
- Regulatory capital is determined by assumptions about dependence and tail behavior.

What This Course Is About (Modeling View)

- **1. Modeling individual default.** Describe whether a firm defaults as a random outcome driven by its underlying financial condition.

- **2. From individual firms to a portfolio.** Aggregate many firms to understand how individual defaults translate into total portfolio losses.

- **3. Comovement and systemic risk.** Capture how firms are jointly affected by common economic factors, leading to correlated defaults.

- **4. Risk management and capital.** Quantify potential losses and determine how much capital is needed to absorb extreme outcomes.

Credit Risk Measurement

Python

Python in This Course

Python for Credit Risk Modelling

In this course, Python is used as a **research environment**, not as a programming course.

Focus:

- implementing quantitative models
- running simulations
- analysing risk distributions
- producing reproducible results

Goal: Use programming as a tool for financial modelling.

Course Philosophy

Modern coding tools and AI can assist with writing code.

However, successful quantitative work still requires:

- understanding model structure
- debugging incorrect implementations
- designing simulations
- interpreting statistical outputs

Key principle:

Programming is a **tool for modelling and analysis**, not an end in itself.

Development Environment Used in This Course

In this course we standardize the working environment to simplify collaboration.

Two main tools will be used:

- **PyCharm** as the primary Python development environment
- **GitHub** as the collaboration and version control platform

Python: Setup

Installing Python

Python must be installed before setting up the development environment.

Step 1: Download Python

- Visit: `https://www.python.org`
- Download the latest stable version of Python
- Recommended: Python 3.x

Step 2: Install Python

- Run the installer
- Select **Add Python to PATH**
- Complete the installation

Verification

Installing PyCharm

PyCharm will be used as the main development environment.

Step 1: Download PyCharm

- Visit: `https://www.jetbrains.com/pycharm`
- Download **PyCharm Community Edition**

Step 2: Install the IDE

Step 3: Create a Project

- Open PyCharm
- Click **New Project**
- Select the installed Python interpreter

Git: Setup

Installing Git

Git is required to interact with GitHub repositories.

Step 1: Download Git

- Visit: `https://git-scm.com`
- Download the installer for your operating system

Step 2: Install Git

- Run the installer
- Accept default configuration options

Verification

```
git --version
```

Creating a GitHub Account

GitHub will be used for collaboration.

Step 1: Create an Account

- Visit: `https://github.com`
- Click **Sign Up**
- Create a username and password

Step 2: Configure Git

After installing Git, configure your identity:

```
git config --global user.name "Your Name"
git config --global user.email "your_email@example.com"
```

Workflow

Structured Projects

Typical structure:

```
credit-risk-project/

 data/
 src/
    simulation.py
    models.py
    utils.py
 notebooks/
 results/
 requirements.txt
 README.md
```

Principle: Separate data, code, analysis, and results.

Virtual Environments

Python environments isolate project dependencies.

Benefits:

- consistent package versions
- fewer compatibility issues
- easier replication of results

Common tools:

- `venv`
- `conda`

Dependencies are stored in

`requirements.txt`

Why Version Control?

Without version control:

- files get overwritten
- changes are difficult to track
- collaboration becomes unreliable

Version control systems record:

- who changed code
- what changed
- when it changed

Standard tool: **Git**

Basic Git Workflow

Typical development workflow:

1. Clone the repository
2. Create a new branch
3. Implement a feature
4. Commit changes
5. Submit a pull request

Benefits:

- changes are documented
- work can be reviewed
- errors can be reverted

Collaboration on GitHub

GitHub provides infrastructure for team development.

Main collaboration tools:

- pull requests
- issue tracking
- code review

Typical workflow:

`feature branch` → `pull request` → `review` → `merge`

Team Assignment Structure

Students work in **teams of two to three**.

Project setup:

- Each student creates an individual branch in the repository: `https://github.com/Zazazzzzz/Credit-Risk-Student-Workspace` (*send an email to be added as a collaborator*)
- Each team creates a shared project file and starts developing a script collaboratively
- Submit an initial test contribution (e.g., small script or file) to understand the collaboration workflow
- Use this setup throughout the course to jointly develop and submit the final assignment

Course Development Workflow

Initial Setup:

```
Install Python  →  Install PyCharm (IDE)  →
    Install Git  →  Create GitHub Account
```

After the setup:

```
Create Project → Write Code → Commit → Push →
                Pull Request
```

Objective: Develop models collaboratively while maintaining reproducible and well-structured research code.

Python: Programming

Core Python Libraries

Key tools used in this course:

NumPy

- vectorized computations
- random number generation

Pandas

- portfolio data handling

SciPy

- probability distributions

Matplotlib / Seaborn

- visualization of results

Programming Focus

The course emphasizes:

- Monte Carlo simulation design
- modular functions
- clear model implementation
- efficient numerical computation

The goal is to translate **financial models into executable code**.

Basic Statistics

Why statistics in credit risk?

- Credit risk models produce **random losses**.
- Risk measures (VaR, ES) are **features of distributions**.
- Dependence drives **clustering** and tail risk.

Statistics → Probability → Simulation

Random Variables: The Core Object

A random variable assigns a number to a random outcome.

Credit risk examples

- Y_i: default indicator
- K: number of defaults
- L: portfolio loss
- X_i: latent (asset/creditworthiness) variable

Default indicator

$$Y_i = \begin{cases} 1 & \text{default} \\ 0 & \text{no default} \end{cases}$$

$$P(Y_i = 1) = p_i$$

Random Variable: Discrete vs. Continuous

Discrete

- takes values in a countable set
- examples: default indicator Y_i
- probabilities: $P(Y = 1) = p, \quad P(Y = 0) = 1 - p$

Continuous

- takes values on a continuous range (intervals)
- examples: returns, latent variable X_i
- $P(X = x) = 0$ for any single value x
- described by densities, e.g. $X \sim \mathcal{N}(\mu, \sigma^2)$

Key Distributions I: Bernoulli, Binomial, Poisson

Bernoulli Distribution (one obligor)	**Binomial Distribution** (number of defaults)	**Poisson Distribution** (rare events)
$Y_i \sim \text{Bernoulli}(p_i)$	$K = \sum_{i=1}^{m} Y_i$	$K \sim \text{Poisson}(\lambda)$
$Y_i \in \{0, 1\}$	$K \sim \text{Binomial}(m, p)$	$\lambda = mp$

Independent defaults $\Rightarrow$ Binomial

Large m, small p $\Rightarrow$ Poisson approximation

Key Distributions I: Probability Mass Functions

Probability Mass Functions

- **Bernoulli:**

$$P(Y_i = y) = p_i^y (1 - p_i)^{1-y}, \quad y \in \{0, 1\}$$

- **Binomial:**

$$P(K = k) = \binom{m}{k} p^k (1 - p)^{m-k}, \quad k = 0, 1, \ldots, m$$

- **Poisson:**

$$P(K = k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \ldots$$

- PMF assigns probability to each possible outcome.

Key Distributions II: Normal Distribution

Why normal matters:

- central building block for factor models and Gaussian copula
- convenient for thresholds via $\Phi^{-1}(\cdot)$

$$X \sim \mathcal{N}(\mu, \sigma^2), \qquad Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

Probability Density Function (PDF):

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Threshold link:

$$p_i = P(X_i \leq d_i) \quad \Rightarrow \quad d_i = \Phi^{-1}(p_i)$$

Key Distributions III: Beta and Gamma

Beta Distribution

- Defined on the interval $[0, 1]$
- Flexible shapes for modelling probabilities
- Useful when the probability itself is uncertain

$$p \sim \text{Beta}(\alpha, \beta)$$

Application: modelling uncertainty in PD.

Gamma Distribution

- Positive-valued distribution
- Flexible skewness and tail behaviour
- Often used for random intensities

$$\lambda \sim \text{Gamma}(k, \theta)$$

Application: default intensity models.

These distributions often appear as mixing distributions in credit risk models.

Mixtures and Overdispersion I

In practice, default probabilities are rarely known exactly.

Idea: model the PD itself as random.

$$p \sim \text{Beta}(\alpha, \beta)$$

Conditional on p, defaults follow

$$K \mid p \sim \text{Binomial}(n, p)$$

This leads to a **Beta–Binomial mixture**:

$$K \sim \text{Beta-Binomial}$$

Implication

- higher variance than a binomial model; more frequent clusters of defaults
- captures **overdispersion in credit portfolios**

Mixtures and Overdispersion II

In some models, defaults are viewed as **arrival events over time**.

Poisson model

$$N \mid \lambda \sim \text{Poisson}(\lambda)$$

where λ is the default arrival intensity.

Allow the intensity to be random

$$\lambda \sim \text{Gamma}(k, \theta)$$

Resulting mixture distribution:

$$N \sim \text{Negative Binomial}$$

Implication

- higher variance than Poisson; more clustered default events
- captures **overdispersion in default arrivals**

Mixtures and Overdispersion III

In practice, defaults are driven by an underlying continuous variable.

Idea: link a continuous distribution to a discrete outcome.

$$X_i \sim \mathcal{N}(0, 1)$$

Default occurs if:

$$Y_i = \mathbf{1}(X_i \leq d_i)$$

Where

- X_i represents a latent variable (e.g. asset return or firm condition)
- Y_i is a binary outcome derived from X_i

Implication

- introduces **randomness** through an underlying continuous driver; provides a foundation for **dependence and clustering** via the latent variable

Expected Value: Long-Run Average

Idea: The expected value describes the long-run average outcome.

$$\mathbb{E}[X]$$

Default Indicator

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\mathbb{E}[Y_i] = p_i$$

Portfolio Expected Loss

$$EL = \sum_{i=1}^{m} w_i \, LGD_i \, p_i$$

Expected loss is an average outcome, not a stress scenario.

Variance and Standard Deviation: Uncertainty Around the Mean

Idea: variance measures dispersion; standard deviation is dispersion in original units.

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big], \qquad \sigma(X) = \sqrt{\mathrm{Var}(X)}$$

Bernoulli example (default indicator):

$$\mathrm{Var}(Y_i) = p_i(1 - p_i)$$

Takeaway: low PD does not imply low uncertainty in portfolio outcomes.

Covariance and Correlation: Dependence

Covariance: joint movement (scale-dependent).

$$\mathrm{Cov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big]$$

Correlation: standardized dependence (-1 to 1).

$$\rho(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)}$$

Credit risk message: dependence $\Rightarrow$ clustering $\Rightarrow$ heavier loss tails.

Cumulative Distribution Function (CDF)

Definition:

$$F_L(x) = P(L \leq x), \quad x \in \mathbb{R}$$

Properties:

- $0 \leq F_L(x) \leq 1$
- $F_L(x)$ is non-decreasing in x
- $\lim_{x \to -\infty} F_L(x) = 0, \quad \lim_{x \to \infty} F_L(x) = 1$

Interpretation:

- $F_L(x)$ gives the probability that the loss does not exceed x
- fully characterizes the distribution of L

Quantiles: From Distributions to Risk Measures

Quantile definition:

$$q_\alpha(L) = \inf\{x \in \mathbb{R} : F_L(x) \geq \alpha\}, \quad \alpha \in (0,1)$$

Interpretation:

- $q_\alpha(L)$ is the smallest loss level such that the cumulative probability reaches α
- quantiles characterize tail behavior without symmetry assumptions

Value at Risk:

$$\mathrm{VaR}_\alpha(L) := q_\alpha(L) = F_L^{-1}(\alpha)$$

First Portfolio Simulation

Before introducing theoretical models, we begin with a simple simulation.

Example portfolio:

- A bank lends **€100,000 each to 10 firms**.
- All firms promise to repay the full amount in one year (no interest, single payment).
- Each firm may default:
  - Firm 1: 1% chance of default
  - Firm 2: 2%
  - ...
  - Firm 10: 10%
- If a firm defaults, it only repays **€50,000**
  $\Rightarrow$ the bank loses €50,000

What Do We Want to Understand?

On average:

- How much loss should the bank expect?

In reality:

- Losses are uncertain
- If we repeat this situation **1,000 times (simulation)**, what does the **distribution of total losses** look like?

Simulation Workflow

Simulation steps:

1. simulate default indicators
2. compute losses for each obligor
3. aggregate portfolio losses
4. repeat for many scenarios
5. analyse the loss distribution

This introduces key concepts used later:

- random variables
- probability distributions
- tail risk

Motivation for Credit Risk Models

The simulation illustrates an important question:

How do we model dependence between defaults?

This motivates the models studied later:

- threshold models
- factor models
- copulas
- portfolio loss distributions

Credit Risk

Introduction

What is Credit Risk?

Credit risk is the risk of a loss arising from the failure of a counterparty to honor its contractual obligations. This subsumes both default risk (the risk of losses due to the default of a borrower or a trading partner) and downgrade risk (the risk of losses caused by a deterioration in the credit quality of a counterparty that translates into a downgrading in some rating system).

Obligor: A counterparty with a financial obligation to another party. For example, a debtor who owes money.
Default: The failure to fulfill a financial obligation, such as repaying a loan or paying interest/coupons on a bond.

Basic Concepts

Some fundamental definitions in this context:

- **Probability of Default (p):** the likelihood that a counterparty will fail to meet their debt obligations over a specific time horizon.
- **Exposure at Default (EAD):** measures the total outstanding that is at risk if the borrower defaults.
- **Loss Given Default (LGD):** the percentage of the total exposure that is expected to be lost in the event of default.
- **Expected Loss (EL):** represents the average or anticipated loss that a financial institution expects to incur over a specified period due to defaults and credit impairments.
- **Unexpected Loss (UL):** the loss that exceeds the average expected loss.

Expected vs. Unexpected Loss

Expected Loss (EL)

$$EL = \sum_{i=1}^{m} w_i \, LGD_i \, p_i$$

- Average loss expected over the time horizon
- Driven by:
  - Probability of default p_i
  - Loss given default LGD_i
  - Exposure weight w_i
- Typically covered by pricing, provisions, and reserves

Unexpected Loss (UL)

- Deviation of actual losses from the expected level
- Arises from:
  - Default uncertainty
  - Correlation between obligors
  - Systematic economic shocks
- Managed using **economic and regulatory capital**

The Importance of Dependence Structure

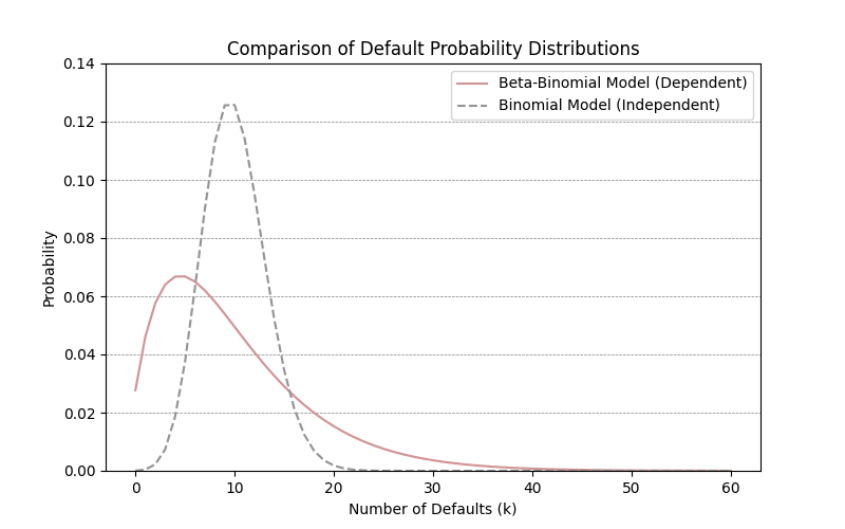


Figure 2: Comparison of the Distribution of Default Numbers of Different Dependence Structures

This plot is a comparison of the loss distribution of a homogeneous portfolio of 1000 loans with a default probability of $p_1 = \cdots = p_{1000} = 1\%$ assuming (i) independent defaults and (ii) a default correlation of 0.5%.

The Importance of Dependence Structure

There are two primary sources of dependence between defaults:

- **Common Factor Dependence:** Defaults may be influenced by shared macroeconomic factors, such as interest rates, GDP growth, or inflation, which impact all obligors simultaneously.
- **Contagion Effects:** The default of one company (e.g., Company A) can directly affect the default probability of another company (e.g., Company B) due to interconnected business relationships, financial linkages, or supply chain dependencies, a phenomenon known as contagion.

OUTCOME: When defaults are correlated, extreme losses become more likely, leading to higher tail risk in the portfolio's loss distribution. In reality, it can cause economic downturns, liquidity crises, or sector-specific shocks.

Default Modeling

Default as a Random Event

Credit risk begins with a simple question:

Does the obligor default over the horizon?

$$Y_i = \begin{cases} 1, & \text{default} \\ 0, & \text{no default} \end{cases}$$

Implications

- Each obligor is reduced to a binary outcome.
- The randomness comes from not knowing in advance which firms will default.
- Portfolio credit risk is built by combining many such indicators.

Threshold Representation of Default

Default rule

$$Y_i = \begin{cases} 1, & \text{if } X_i \leq d_i, \\ 0, & \text{if } X_i > d_i \end{cases}$$

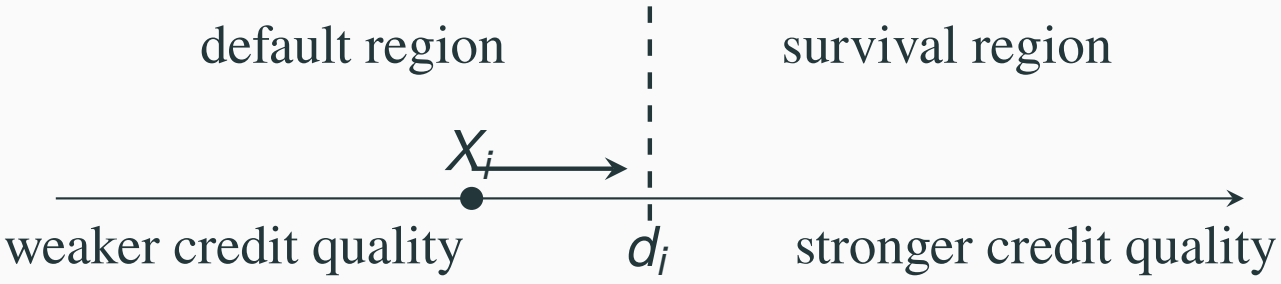
- X_i: latent variable measuring the obligor's credit quality
- d_i: default threshold
- Default occurs when credit quality falls below the threshold

Interpretation

-**High** X_i: firm remains healthy

- **Low** X_i: firm is under distress

- **Threshold crossing**: default event



Structural Model

Merton's Model

Merton's model (1974) serves as the prototype of all firm value models. Consider a firm with a stochastic asset value V_t, financing itself through equity (i.e., by issuing shares) and debt.

- The firm's debt consists of a single zero-coupon bond with a face (or nominal) value B and maturity T.
- Let S_t and B_t denote the values of equity and debt, respectively, at time $t \leq T$.
- The firm's total value satisfies the relationship:

$$V_t = S_t + B_t, \quad 0 \leq t \leq T$$

- Default occurs if the firm fails to meet its payment obligations to debt holders, which can only happen at time T.

Merton's Model

Possible Cases at T:

a. $V_T > B$:
   - The debtholders receive B.
   - The shareholders receive the residual value $S_T = V_T - B$, and there is no default.

b. $V_T \leq B$:
   - The firm cannot meet its financial obligations.
   - Shareholders hand over control to the bondholders, who liquidate the firm.
   - In this case, $B_T = V_T$ and $S_T = 0$.

In summary, we obtain:

$$S_T = (V_T - B)^+$$

$$B_T = \min(V_T, B) = B - (B - V_T)^+$$

Merton's Model

The Asset Value Process:

- It is assumed that the asset value V_t follows a geometric Brownian process of the form:

$$dV_t = \mu_V V_t \, dt + \sigma_V V_t \, dW_t \tag{1}$$

  for constants $\mu_V \in \mathbb{R}$, $\sigma_V > 0$, and a standard Brownian motion $(W_t)_{t \geq 0}$.
- The solution at time T of the stochastic differential equation (1) is given by:

$$\ln V_T = \ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T, \tag{2}$$

$$V_T = V_0 \exp\left(\left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T\right) \tag{3}$$

Merton's Model

Default Probability:

From equation (2), we know $\ln V_T \sim N\left(\ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T, \sigma_V^2 T\right)$.

The default probability is therefore:

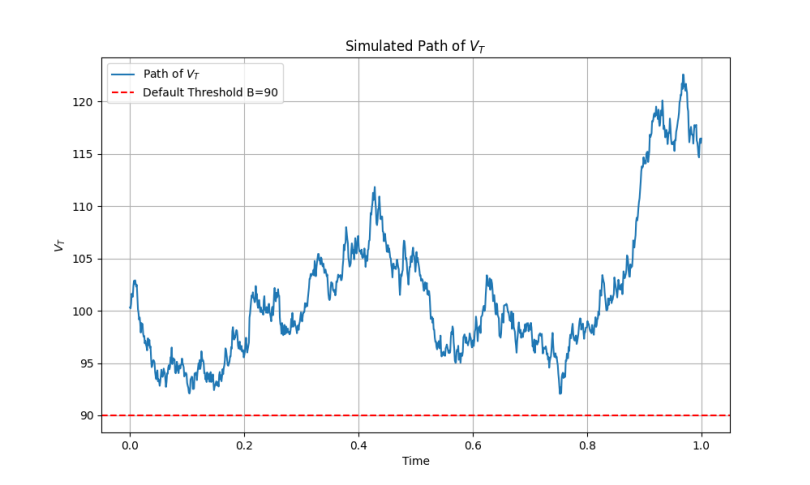
$$\begin{aligned} P(V_T \leq B) &= P(\ln V_T \leq \ln B) \\ &= \Phi\left(\frac{\ln B - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}\right). \end{aligned} \tag{4}$$

The Merton model is equivalent to the threshold model (X, d) with

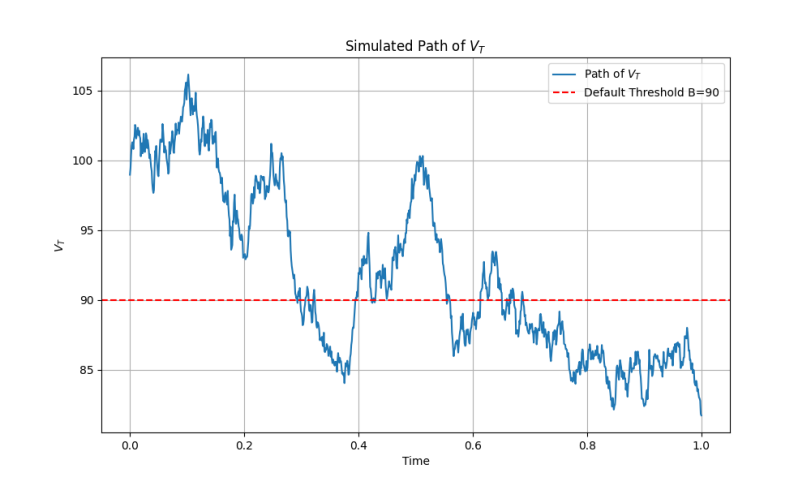
$$X := \frac{\ln V_T - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}},$$

$$d := \frac{\ln B - \ln V_0 - \left(\mu_V - \frac{1}{2}\sigma_V^2\right)T}{\sigma_V\sqrt{T}}.$$

Merton's Model



(a) Path without default.



(b) Path with default

Figure 3: Comparison of default paths for V_T.

Task 1: Simulate default path under Merton's model

To simulate Figure 3a.

Parameters: `V_0 = 100, mu_V = 0.05, sigma_V = 0.2, T = 1.0, B = 90, N = 1000, dt = T / N, np.random.seed(42)`.

Portfolio

Portfolio Setting

- We consider a portfolio of m obligors and fix a time horizon T.
- The random vector $\mathbf{Y} = (Y_1, \ldots, Y_m)'$ is a vector of default indicators for the portfolio.
- The marginal default probabilities are denoted by $p_i = P(Y_i = 1)$, where $i = 1, \ldots, m$.
- The joint probability is $p(\mathbf{y}) = P(Y_1 = y_1, \ldots, Y_m = y_m), \quad \mathbf{y} \in \{0, 1\}^m$

Task 2: Simulate default outcome in a portfolio

Simulate 1000 trials of a portfolio consisting of 100 obligors, each with a 1% probability of default, assuming the default events are independent. Then, plot the density of the number of defaults observed in the simulations.

Parameters:

`N = 1000, P(Y_i) = 1%, m = 100, np.random.seed(42).`

Portfolio Setting

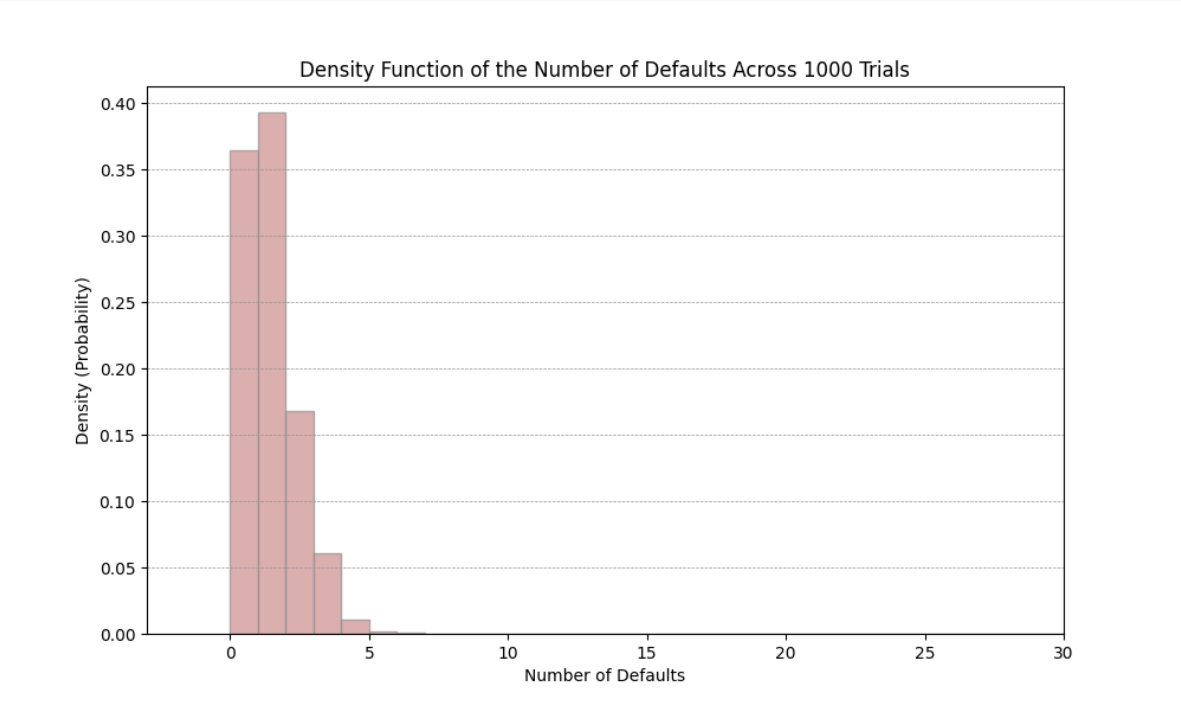


Figure 4: Simulation of Default Outcomes of a Portfolio With No Interdependence: Homogeneous obligors

Dependency Structure in a Portfolio

In the same setting, we derive the **default correlation.** Noting that

$$\mathrm{var}(Y_i) = \mathbb{E}(Y_i^2) - p_i^2 = \mathbb{E}(Y_i) - p_i^2 = p_i - p_i^2,$$

we obtain, for firms i and j with $i \neq j$, the formula

$$\rho(Y_i, Y_j) = \frac{\mathbb{E}(Y_i Y_j) - p_i p_j}{\sqrt{(p_i - p_i^2)(p_j - p_j^2)}}. \tag{5}$$

Dependency Structure in a Portfolio

Task 3: The dependency structure of the obligors

Y_i for $i = 1, 2$ is the default indicator during a time period for two companies. Now, assume we have the following joint probabilities:

$$P(Y_1 = 0, Y_2 = 0) = 0.85 \quad P(Y_1 = 1, Y_2 = 1) = 0.05$$
$$P(Y_1 = 1, Y_2 = 0) = 0.07 \quad P(Y_1 = 0, Y_2 = 1) = 0.03$$

- Compute the marginal probabilities of default for Y_1 and Y_2. And the default correlation.
- Compute the joint probability of default, that is $P(X_1 \leq d_1,\ X_2 \leq d_2)$ (threshold model), that is the $C(p_1, p_2)$.

Marginal and Joint Distribution

The given probabilities describe the *joint distribution* of (Y_1, Y_2). That is, they describe the probabilities of all possible combinations of default and non-default for the two obligors.

The *marginal default probability* of one obligor is obtained by summing over all outcomes of the other obligor.

For Y_1, we have

$$p_1 = P(Y_1 = 1) = P(Y_1 = 1, Y_2 = 0) + P(Y_1 = 1, Y_2 = 1).$$

Therefore,

$$p_1 = 0.07 + 0.05 = 0.12.$$

For Y_2, we have

$$p_2 = P(Y_2 = 1) = P(Y_1 = 0, Y_2 = 1) + P(Y_1 = 1, Y_2 = 1).$$

Default Correlation

The *default correlation* measures the dependence between the default events Y_1 and Y_2. It is defined as according to (5):

$$\frac{P(Y_1 = 1, Y_2 = 1) - P(Y_1 = 1)P(Y_2 = 1)}{\sqrt{P(Y_1 = 1)(1 - P(Y_1 = 1))P(Y_2 = 1)(1 - P(Y_2 = 1))}}.$$

Substituting the values gives

$$\rho(Y_1, Y_2) = \frac{0.05 - (0.12)(0.08)}{\sqrt{0.12(1 - 0.12) \cdot 0.08(1 - 0.08)}}.$$

Hence,

$$\rho(Y_1, Y_2) = \frac{0.0404}{0.0883} \approx 0.457.$$

Joint Default Probability

The *joint probability of default* is

$$P(Y_1 = 1, Y_2 = 1) = 0.05.$$

Using the marginal default probabilities

$$p_1 = P(X_1 \leq d_1), \qquad p_2 = P(X_2 \leq d_2),$$

the copula representation gives

$$P(X_1 \leq d_1,\ X_2 \leq d_2) = C(p_1, p_2).$$

Thus,

$$C(p_1, p_2) = P(Y_1 = 1, Y_2 = 1) = 0.05.$$

Dependency Structure in a Portfolio

We consider a special case: **an exchangeable default model**.

- It is common to group obligors together to form homogeneous groups. This corresponds to the mathematical concept of exchangeability.
- Note that this permits a simple notation for default probabilities:

$$\pi_k := P(Y_{i_1} = 1, \ldots, Y_{i_k} = 1), \quad \{i_1, \ldots, i_k\} \subset \{1, \ldots, m\}, \tag{6}$$

$$\pi := \pi_1 = P(Y_i = 1), \quad i \in \{1, \ldots, m\}. \tag{7}$$

- In the exchangeable case, the default correlation is given by

$$\rho_Y := \rho(Y_i, Y_j) = \frac{\pi_2 - \pi^2}{\pi - \pi^2}, \quad i \neq j.$$

Default correlation and asset correlation

- The probability of default for an obligor is given by:

$$P(Y_i = 1) = P(X_i \leq d_i) = F_{X_i}(d_i).$$

- Default correlation: For given default probabilities, the correlation between defaults $\rho(Y_i, Y_j)$ is determined by $\mathbb{E}(Y_i Y_j)$ according to (5). In a threshold model, $\mathbb{E}(Y_i Y_j) = P(X_i \leq d_i, X_j \leq d_j)$, so the default correlation depends on the joint distribution function of X_i and X_j.
- Asset correlation: Since the critical variables are often interpreted in terms of asset values, the correlation of the critical variables X_i and X_j is often referred as asset correlation.

Why Asset Correlation Is Not Default Correlation

Even if X_i and X_j are strongly correlated, defaults only occur in the lower tail:

$$X_i \leq d_i, \qquad X_j \leq d_j.$$

Therefore, default correlation depends not only on $\rho(X_i, X_j)$, but also on the default thresholds d_i, d_j, or equivalently the default probabilities p_i, p_j.

$$P(Y_i = 1, Y_j = 1) = P(X_i \leq d_i,\ X_j \leq d_j).$$

So the same asset correlation can lead to different default correlations if the marginal default probabilities are different.

$$\rho(X_i, X_j) \neq \rho(Y_i, Y_j)$$

Dependency Structure in a Portfolio

Asset Correlation: Measures the correlation between the latent variables X_i. It is a parameter of the factor model and directly controls the dependence structure of the X_i.

Default Correlation: Measures the correlation between default events Y_i. It is influenced by the asset correlation and the marginal probability of default p_i.

Factor Models

Why Factor Models?

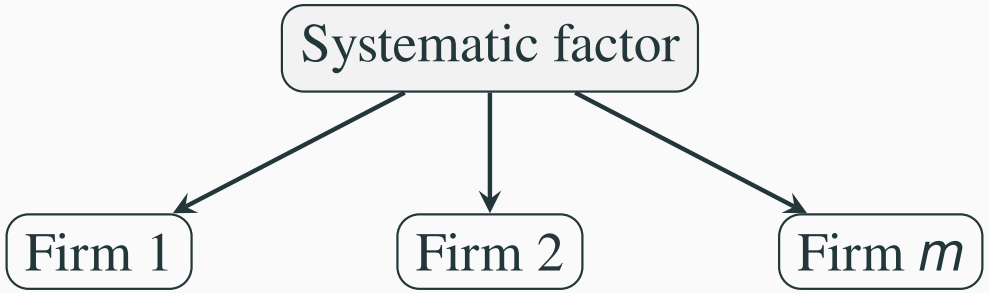
Economic reality

- Firms are exposed to the same macroeconomic environment
- Economic shocks affect many companies simultaneously
- Defaults therefore tend to **cluster**

Defaults are not independent events.

Goal of factor models

- Introduce **systematic risk factors**
- Allow many obligors to respond to the same shock
- Generate **correlated defaults** in a tractable way



One systematic shock → many obligors

Factor Models

Latent Variable Factor Models

Latent Variable Approach

For an obligor i, the **critical variable** X_i is modeled as:

$$X_i = \rho_i \tilde{F}_i + \sqrt{1 - \rho_i^2}\epsilon_i, \tag{8}$$

where:

- X_i: Critical variable for obligor i. A lower value of X_i indicates higher credit risk.

- $\tilde{F}_i$: Common factor (systematic risk), typically assumed to follow a normal distribution.

- ρ_i: Sensitivity of obligor i to the common factor (asset correlation).

- ϵ_i: Idiosyncratic component (obligor-specific risk), assumed to follow a standard normal distribution ($\epsilon_i \sim N(0, 1)$) and independent of $\tilde{F}_i$ and other ϵ_j.

Systematic Factors

Now, let us consider **systematic factors** $\tilde{F}_i$.

$$\tilde{F}_i = \mathbf{a_i'F} = a_{i1}F_1 + a_{i2}F_2 + \cdots + a_{ip}F_p \tag{9}$$

- $\tilde{F}_i$: This is the systematic risk component for asset i. $\text{var}(\tilde{F}_i) = 1$.

- $\mathbf{a_i}$: This is a vector of factor loadings for asset i. It determines how much asset i is influenced by each of the common factors in the vector $\mathbf{F}$.

- $\mathbf{F}$: This is a vector of common factors. These factors could represent broad economic variables such as market conditions, industry performance, etc. that affect all assets in the portfolio.

- $F \sim N_p(0, \Omega)$: The common factor vector F follows a multivariate normal distribution with a mean vector of zeros and a covariance matrix Ω.

Default as a Threshold Event

Default rule

$$Y_i = \begin{cases} 1, & \text{if } X_i \leq d_i \\ 0, & \text{otherwise} \end{cases}$$

Link to probability of default

$$P(X_i < d_i) = p_i$$

- X_i : latent credit quality of obligor i
- d_i : default threshold
- p_i : probability of default

Interpretation

- Lower values of X_i correspond to weaker credit quality
- Default occurs when the threshold is crossed
- The threshold d_i is chosen so that

$$P(X_i < d_i) = p_i$$

Asset Correlation

The **asset correlation** $\rho(X_i, X_j)$ can be derived as following:

First, we compute covariance between X_i and X_j. Since the idiosyncratic terms ϵ_i and ϵ_j are independent, they do not contribute to the covariance. And $\text{cov}(\tilde{F}_i, \tilde{F}_j) = a_i' \text{var}(F) a_j = a_i' \Omega a_j$:

$$\text{cov}(X_i, X_j) = \text{cov}(\rho_i \tilde{F}_i + \sqrt{1 - \rho_i^2}\epsilon_i,\ \rho_j \tilde{F}_j + \sqrt{1 - \rho_j^2}\epsilon_j)$$

$$= \rho_i \rho_j \text{cov}(\tilde{F}_i, \tilde{F}_j) = \rho_i \rho_j a_i' \Omega a_j$$

Thus, the asset correlation is:

$$\rho(X_i, X_j) = \frac{\text{cov}(X_i, X_j)}{\sqrt{\text{var}(X_i)\text{var}(X_j)}}$$

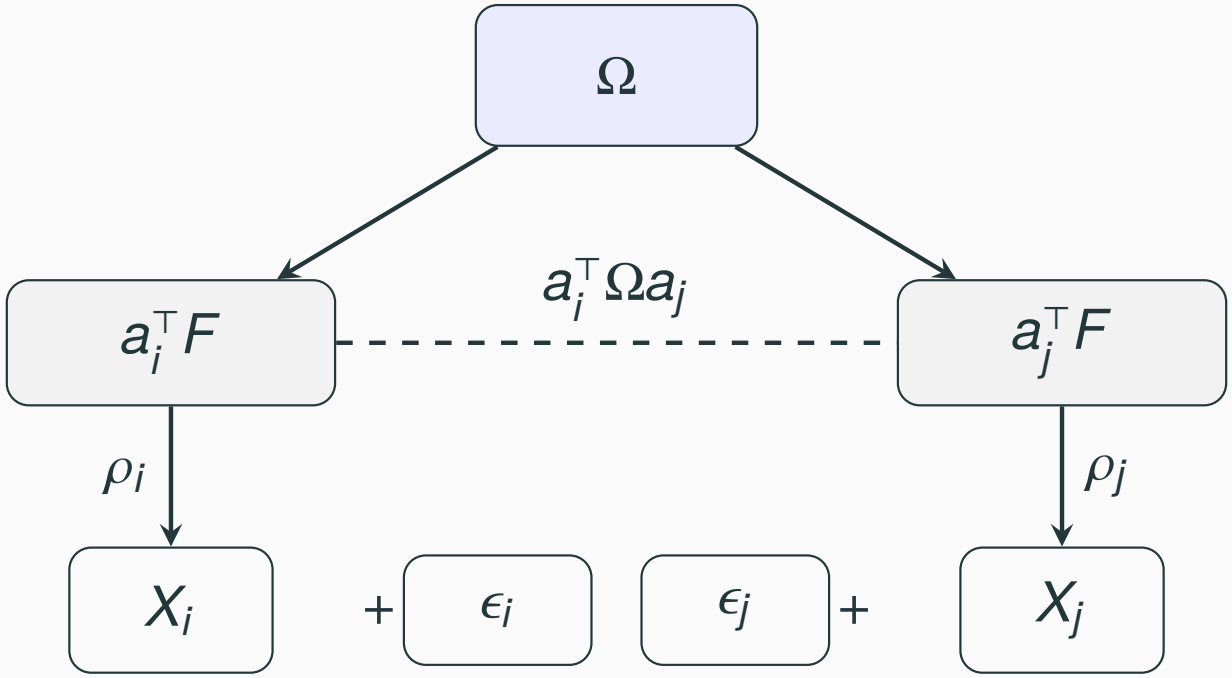
Given that $\text{var}(\tilde{F}_i) = 1$ and therefore, X_i and X_j have unit variance,

$$\rho(X_i, X_j) = \text{cov}(X_i, X_j) = \rho_i \rho_j a_i' \Omega a_j \tag{10}$$

Asset Correlation

Key intuition

Dependence between firms arises share exposure to the same systematic risk factors.



One-Factor Model

If we assume there is only one **systematic factor F** for all the obligors:

$$\tilde{F}_i = F_1 = F_2 = \cdots = F_m = F \tag{11}$$

For obligor i's creditworthiness:

$$X_i = \rho_i F + \sqrt{1 - \rho_i^2}\, \epsilon_i \tag{12}$$

where F is a common factor (e.g., "the economy"). $F \sim N(0, \sigma^2)$.

Real-World Complexities

- **Oil vs Tech**: Both affected by GDP, but differently by oil prices
- **Geographies**: US and German firms react differently to EUR/USD rates

Homogeneous Portfolio

Now, we consider a special case of **one-factor model**, where $\tilde{F}_i = F$ for all i. Moreover, a **homogeneous portfolio**, where the obligors have the same asset correlation, denoted by $\rho_i = \rho$. The critical variable X_i takes the form:

$$X_i = \rho F + \sqrt{1 - \rho^2}\epsilon_i \tag{13}$$

where $F \sim N(0, 1)$ represents the common systematic risk factor, and $\epsilon_i \sim N(0, 1)$ are idiosyncratic risk terms that are independent of F and of each other. Since both F and ϵ_i are standard normal, it follows that $X_i \sim N(0, 1)$.

An obligor i is considered to default if X_i falls below a threshold d_i, which corresponds to their default probability:

$$P(X_i < d_i) = p_i, \quad \text{where } d_i = \Phi^{-1}(p_i).$$

Homogeneous Portfolio

Task 4: Default behaviors of obligors

Simulating the default behavior of a portfolio of 1000 obligors, each with a default probability of 1%. Two scenarios are considered:

- independent defaults
- correlated defaults, where defaults are influenced by a systemic factor with ρ of 0.25

Simulate both scenarios for 2000 times and visualize the result through histograms.

Parameters:

`N = 2000, P(Y_i) = 1%, m = 1000, np.random.seed(42).`

Homogeneous Portfolio

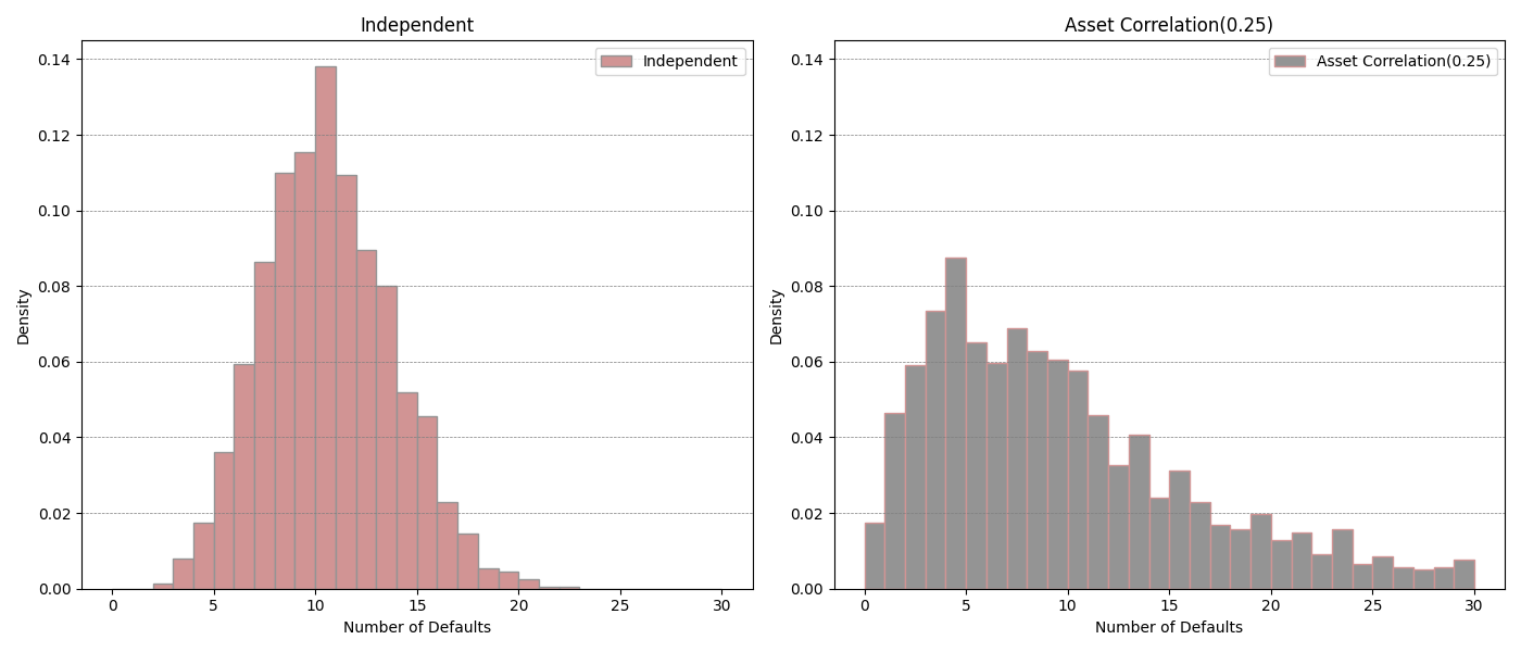


Figure 5: Simulation of the Default Numbers of different dependency structures

Independent defaults: The probabilities will be tightly clustered around the mean (10 defaults). Correlated defaults: The probabilities will have a *fatter tail*, indicating a higher likelihood of extreme losses.

Task 5: Estimate the probability of k defaults*

Under the same context, compute the probabilities for each possible number of defaults and plot the probabilities for both scenarios to compare the distributions.

HINT: The probabilities of the independent defaults can be computed using the binomial probability density function:

$$P(k) = \binom{1000}{k}(0.01)^k(0.99)^{1000-k}.$$

The probabilities of the correlated default can be estimated empirically from the simulation results by counting the frequency of each number of defaults.

Homogeneous Portfolio

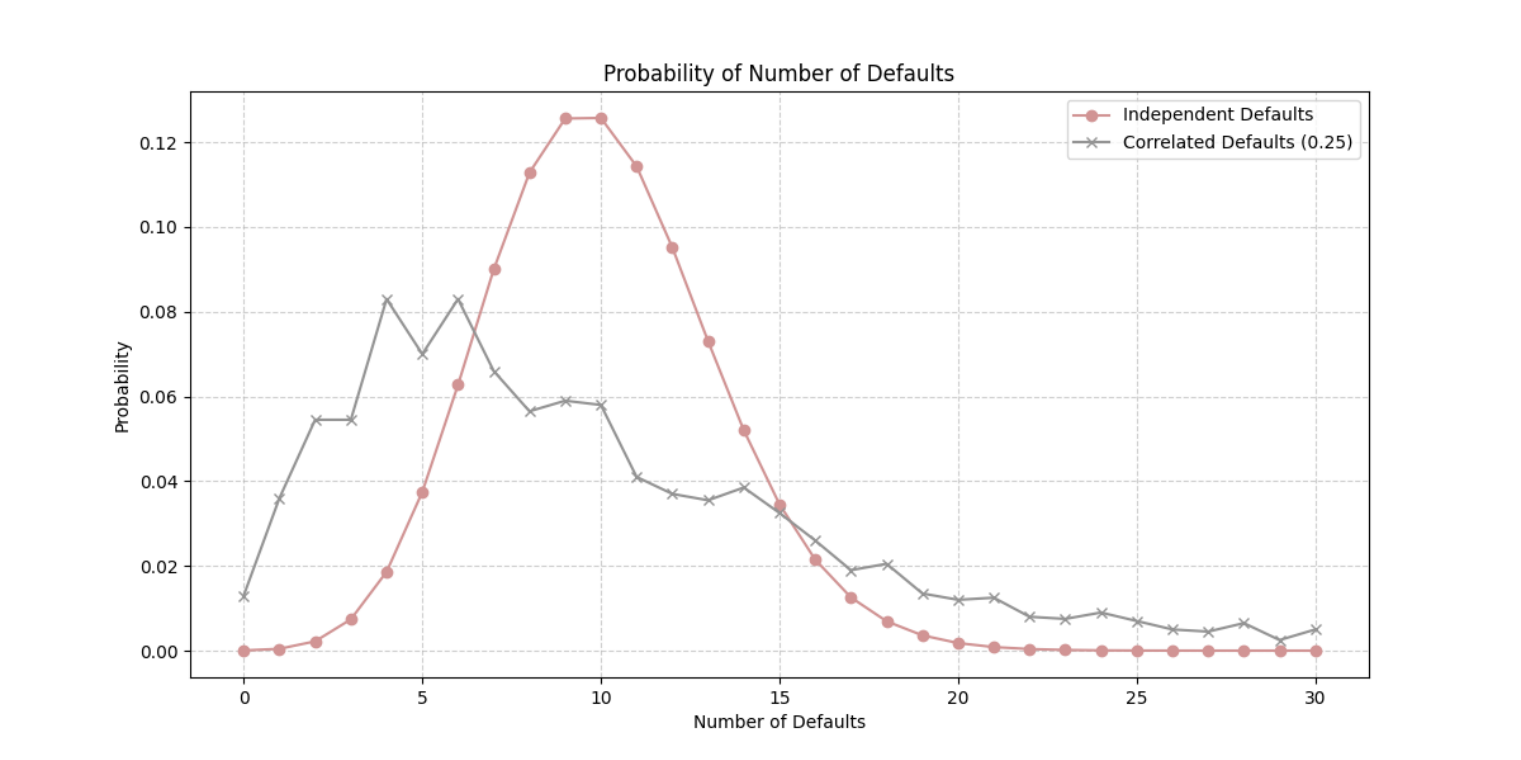


Figure 6: Estimated probability of default numbers